

Graded Support Audit Status Note

March 21, 2026

Status

This note records the current status of the graded-support mesoscape sandbox after the first constraint probe and the first move audit.

1 Purpose

The graded-support sandbox was introduced to replace binary support ontology with graded support commitment

$$\sigma_i \in [0, 1]$$

and graded interface commitment

$$w_{ij} \in [0, 1].$$

The guiding motivation was that the earlier mesoscape evidence pointed toward niches that wanted more than zero support but less than a full stable subsystem, so the binary state space was too coarse. The graded sandbox therefore tested whether persistent partial carriers could be represented directly rather than being approximated by repeated birth/retire cycling.

The present note records what the first graded run and its audits actually showed.

2 Empirical result of the graded-support run

The graded-support run did *not* collapse into the proto-generalization failure mode. It did not produce a star centered on a single parent. Instead it produced a distributed partial-support network with all 8 nodes active, but only two fully committed and the rest held at partial commitment. In the final state,

$$\sigma = (1, 1, 0.35, 0.35, 0.35, 0.35, 0.35, 0.35),$$

so nodes 0 and 1 are fully instantiated while nodes 2 through 7 remain persistent partial carriers. The final active graph is distributed rather than fan-like. :contentReference[oaicite:0]index=0

At the run level, the accepted move counts were:

Move type	Count
raise_support	6
edge_up	20
edge_down	4
lower_support	0

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So the run naturally separated into two phases:

1. support seeding / partial-carrier creation,
2. later interface retuning at fixed mean support.

After all eight sites became active, the mean support commitment plateaued at

$$\bar{\sigma} = 0.5125,$$

and the later dynamics were almost entirely edge adjustments rather than further support raising or support lowering. :contentReference[oaicite:2]index=2

3 Constraint-probe reading

The graded constraint probe gave a useful constraint-native interpretation of the same run.

Its main read was:

- finite bandwidth is the best candidate for why support mass spreads across many partial carriers rather than concentrating into a few full ones,
- no-refolding makes the resulting late-time support geometry sticky,
- no-forgetting preserves the partial carriers once they exist,
- no-signalling is most visible in the late move flow shifting into local interface retuning rather than further support creation.

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Numerically, the probe found:

- 6 partial nodes,
- partial support mass = 2.1,
- total support mass = 4.1,
- diffuse partial fraction ≈ 0.512 ,
- finite-bandwidth crowding proxy = 18.0,
- first full occupancy at step 80,
- post-full partial persistence fraction = 1.0,
- post-full full-node stability fraction = 1.0,
- late edge Jaccard mean = 1.0,
- late sigma L^1 drift = 0.0,
- late move flow 100% edge-up, 0% support raises, 0% support lowers.

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So at the level of global regime-shape, the graded-support ontology appears compatible with the four available constraint pressures without introducing any new ad hoc pressure term. :contentReference[oaicite:5]index=5

4 Move-audit result

The graded move audit sharpens that picture significantly.

It shows that the graded-support run is *not* actually being driven, move by move, by finite bandwidth. Instead:

1. every accepted move is dominated in magnitude by the expression term,
2. early accepted `raise_support` moves are mainly constraint-filtered by no-signalling,
3. late accepted `edge_up` moves are mainly constraint-filtered by no-refolding,
4. finite bandwidth is not the operative local selector in the present sandbox.

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The audit’s summary is therefore:

selection is expression-led, constraint-filtered, not constraint-led.

Early regime

For accepted `raise_support` moves, the audit shows large positive

$$dE_{\text{expr}}$$

while the finite-bandwidth term is essentially zero and no-signalling is the main active penalty among the constraint terms. :contentReference[oaicite:7]index=7

Late regime

For accepted late `edge_up` moves, raw expression is slightly negative, but total

$$dE_{\text{expr}}$$

remains positive because the sandbox’s graded edge-up bonus and related adjustment structure keep it positive. The late-regime summary explicitly says:

- late regime is interface-dominated rather than support-raising dominated,
- no-refolding appears to be the strongest late-time lawful selector.

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So the late interface-retuning phase is real, but the reason those moves keep winning is not simply that the four lawful pressures demand them. It is that the expression-side scoring still favors them.

5 What this means

The graded-support ontology itself looks promising. It solved a real representation problem:

- it prevented collapse into a one-parent fan,
- it represented persistent partial carriers directly,

- it allowed later dynamics to shift into interface retuning instead of binary support churn.

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But the move audit shows that the current sandbox scoring is still too permissive on the expression side.

In particular:

1. finite bandwidth explains the *global viability* of diffuse partial support better than it explains *local move selection*,
2. no-signalling mainly filters early support-raising,
3. no-refolding mainly filters late interface retuning,
4. accepted late **edge_up** moves are still being sustained by positive expression-side adjustments even when raw local expression gain is slightly negative.

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So the sandbox has solved the ontology problem better than it has solved the lawful move-selection problem.

6 Main conclusion

The current graded-support program should therefore be described as follows:

Graded support is a promising replacement for binary support ontology, and it appears compatible with the four allowed constraint pressures. However, in the present sandbox the local move selector remains expression-led. Finite bandwidth is not yet functioning as a real local selector, and late edge-up acceptance is likely being sustained by the sandbox's expression-side bonuses rather than by raw lawful necessity alone.

This is not a reason to abandon graded support. It is a reason to be more disciplined about how graded expression is scored.

7 What comes next

The next task is *not* to add a fifth pressure. Only the existing constraint pressures are allowed.

The next task is instead:

rewrite graded expression so that it reflects the four lawful constraint pressures more faithfully, rather than overriding them with a generous positive adjustment layer.

More concretely, the next code-facing question is:

Can graded expression be reformulated so that finite bandwidth, no-signalling, no-refolding, and no-forgetting determine graded move selection more directly, instead of serving mainly as small filters on top of an already generous expression bonus structure?

That is the real next step.

One-sentence summary

The graded-support ontology now looks substantively better than the binary one, but the present sandbox still chooses moves in an expression-led way: the four lawful constraint pressures shape the regime, yet finite bandwidth is not yet a genuine local selector and late edge-up moves remain too easy because expression-side bonuses keep them positive. :contentReference[oaicite:11]index=11
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